

Dynamics of virtual electrode-induced scroll-wave reentry in a 3D bidomain model

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Sambelashvili, Aleksandre, and Igor R. Efimov. Dynamics of virtual electrode-induced scroll-wave reentry in a 3D bidomain model. *Am J Physiol Heart Circ Physiol* 287: H1570–H1581, 2004; 10.1152/ajpheart.01108.2003.—Functional reentry in the heart can be caused by a wave front of excitation rotating around its edge. Previous simulations on the basis of monodomain cable equations predicted the existence of self-sustained, vortex-like wave fronts (scroll waves) rotating around a filament in three dimensions. In our simulations, we used the more accurate bidomain model with modified Beeler-Reuter ionic kinetics to study the dynamics of scroll-wave filaments in a $16 \times 8 \times 1.5$ -mm slab of ventricular tissue with straight fibers. Wave fronts were identified as the areas with inward current. Their edges represented the filaments. Both transmural and intramural reentries with I- and U-shaped filaments, respectively, were obtained by the S1-S2 point stimulation protocol through the virtual electrode-induced phase singularity mechanism. The filaments meandered along elongated trajectories and tended to attach to the tissue boundaries exposed to air (no current flow) rather than to the bath (zero extracellular potential). They completely detached from electroporated (zero transmembrane potential) boundaries. In our simulations, the presence of the bath led to generation of only U-shaped filaments, which survived for the 1.5-mm-thick slab but not for the slabs of 0.5- or 3-mm thicknesses. Thus boundary conditions may be another determinant of the type and dynamics of reentry.

virtual anodes

REENTRY IS BELIEVED TO BE a cause of cardiac tachyarrhythmias. One of the various forms of reentry is functional reentry, which can occur in structurally normal myocardium and is usually associated with a sustained vortex-like electrical activity. A possible way to elicit such reentry is to apply an external electrical stimulus to the tail of a propagating wave of excitation (31, 38, 76, 79). In this situation, the stimulus-induced excitation will first spread into the already recovered area of the tissue, and at the same time, as more and more tissue recovers, it will gradually reenter into the initial refractory region. Hence, the wave front of activation at the surface of the tissue will not form a closed loop but rather will be broken by the still refractory area (78). At every particular instant of time, the wave will be spiraling around its edges, which are called wavebreaks, points of phase singularity (81, 83), or critical points (32). The same picture extended to the three dimensions (3D) of the ventricular wall will represent a scroll-like surface of the wave front rotating around the lines of phase singularity termed scroll-wave filaments (49, 82, 85).

Extensive research on critical points in two dimensions (2D) and scroll-wave filaments in 3D has been done theoretically and computationally on the basis of the monodomain model of cardiac muscle described by a cable equation with active

cellular kinetics (10, 29, 43, 55, 57, 59, 85). 2D simulations demonstrate that the phase singularity points do not stay at one location but rather meander along complex trajectories that constitute the so-called core of the reentry (16, 17, 28, 86). The shape of these trajectories strongly depends on the parameters of the model (86, 87). Efimov et al. (26) showed that as the excitability of the cells increased, the initially circular core of the reentry changed to epicycloidal, cycloidal, and hypocycloidal, finally becoming extremely elongated (nearly linear) with sharp turns at the ends. The linear core corresponded to the parameter values for a normal myocyte and was observed experimentally at the surface of the heart (3, 15, 16). On the other hand, strong suppression of excitability caused shortening of the wave front and self-termination of the reentrant spiral wave. Another factor affecting the tip trajectory is the wavelength, which can be changed by altering calcium channel dynamics (26, 41).

Inclusion of 3D adds more challenge to the problem of describing the shape and dynamics of scroll waves and filaments of phase singularity. It has been shown theoretically that three types of filaments are topologically possible: I-, U- and O-shaped (58), depending on where the ends of the filaments are located (49, 57, 85). No filament can terminate inside normal heart tissue, i.e., a filament needs a boundary for attachment, unless it is a self-connected filament (60). Two counterrotating scroll waves can collide and annihilate (29, 53, 57, 83) with the total topological charge being always conserved (54). An opposite process of breakup can also occur when an intramural filament breaks through to the surface creating two counterrotating rotors (11, 29, 42, 53). The rotation of the layers of muscle fibers inside the ventricular wall may play an important role in generation and evolution of a scroll wave (56), force a filament to align with the local fiber orientation (8), or cause its destabilization (29). Heterogeneity of the medium (36, 75), as well as boundary conditions and structure (30, 64) can also be the reason for filament drift, meandering, or destabilization. However, the specific behavior may strongly depend on the chosen parameters of the simulation (11, 26, 29).

Recent studies (23, 25, 68, 84) have revealed the possibility of phase singularity generation even in tissue that does not have any initial gradient of excitability. This possibility is on the basis of the concept of virtual electrodes (VEs), which are adjacent areas of opposite polarization (11, 63) created during application of an electrical stimulus to the heart (70). Figure 1 shows a typical pattern of adjacent hyperpolarization (blue) and depolarization (red) corresponding to the stimulation of the tissue with a cathodal current. If the tissue was initially refrac-

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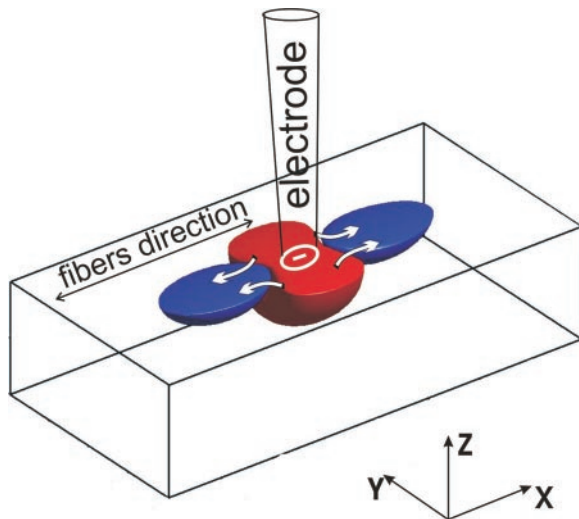


Fig. 1. Schematic diagram of the model. A cathodal current stimulus is applied to the center of the upper surface of the slab of myocardial tissue that was in a relative refractory state before the stimulation. As a result, a characteristic stimulus-induced virtual electrode polarization pattern develops with the central depolarized region of virtual cathode (red) and the hyperpolarized regions of virtual anode (blue). The latter deexcite the tissue providing the substrate for the wave front propagation as illustrated by the arrows.

tory, the hyperpolarized areas of the virtual anodes (VAs) located at some distance from the electrode would be deexcited (7, 77), i.e., become excitable again, and would provide an initial substrate for the spread of activation from the depolarized central area of the virtual cathode (VC) on withdrawal of the pulse (as indicated by the arrows). By the time activation reaches the boundary of shock-deexcited VA areas, the surrounding tissue would recover. Hence, the activation would spread to the sides and later reenter back into the central region, completing the reentrant loop. Lines of phase singularity would be created at the junction of initially depolarized, hyperpolarized, and refractory areas. The described scenario of reentry induction is called the VE-induced phase singularity mechanism (6, 23, 47, 48). It reflects the idea that a single stimulus can itself be a cause of the gradient of excitability and, at the same time, a trigger for the wave of break excitation.

Thus reentry can be induced by VEs. However, VEs themselves are the result of intrinsic heterogeneity of cardiac tissue. The latter is represented in the bidomain model as two interpenetrating extra- and intracellular domains with different conductivities along and across the direction of the fibers, coupled via membrane resistance (33, 50, 73). The bidomain model allowed predicting theoretically that unequal anisotropy of the two domains determined the existence of adjacent areas of positive and negative polarization for any type of electric stimulus applied externally (63, 70). The VE phenomenon has been found and studied in numerous experiments using voltage-sensitive dye techniques (24, 44, 51, 80).

Several works investigated the dynamics of the VE-induced-phase singularities in 2D bidomain models. Skouibine et al. (72) demonstrated that the majority of the phase singularities created by the shock quickly annihilate and only a few sustain the reentry. Bray et al. (12) classified phase singularities on the basis of how their separation changes with time. However, these bidomain studies are limited to the 2D case because the 3D bidomain model is more computationally expensive.

In this study, we aimed to investigate the dynamics of VE-induced filaments in a 3D active bidomain model as well as to link it to the conditions at the boundaries of the cardiac muscle.

Table 1. Parameters used in simulations

Parameter	Abbreviation	Value
Membrane capacitance	C_m	$0.01 \mu\text{F}/\text{mm}^2$
Surface-to-volume ratio	β	300 mm^{-1}
Intracellular conductivity along the fiber	σ_{ix}	$0.1863 \text{ mS}/\text{mm}$
Intracellular conductivity across the fiber	$\sigma_{iy,z}$	$0.0186 \text{ mS}/\text{mm}$
Interstitial conductivity along the fiber	σ_{ex}	$0.1863 \text{ mS}/\text{mm}$
Interstitial conductivity across the fiber	$\sigma_{ey,z}$	$0.0745 \text{ mS}/\text{mm}$
Transmembrane potential at resting state	V_{rest}	-84.54 mV
S1, stimulus strength	I_{01}	-0.5 mA
S1, stimulus duration	d_{01}	2.0 ms
S2, stimulus strength	I_{02}	-3.0 mA
S2, stimulus duration	d_{02}	20.0 ms
S1–S2, coupling interval	CI	88 ms
Typical tissue size	LL	$16 \times 8 \times 1.5 \text{ mm}^3$
Space step along the fiber	Δx	0.1 mm
Space step across the fiber	Δy	0.05 mm
Time step	Δt	$10 \mu\text{s}$

METHODS

A thin rectangular slab of cardiac ventricular tissue with straight fiber geometry (Fig. 1) was modeled on the basis of the bidomain approach (35, 73). The typical size of the slab was $16 \times 8 \times 1.5 \text{ mm}^3$, although these dimensions could vary from simulation to simulation. The following two bidomain equations describe the electrical behavior of the tissue

$$\nabla \cdot (\hat{\sigma}_i \nabla V_m) + \nabla \cdot [(\hat{\sigma}_i + \hat{\sigma}_e) \nabla \phi_e] = -I_0$$

$$\nabla \cdot (\hat{\sigma}_e \nabla \phi_e) = -I_m - I_0$$

where V_m and ϕ_e are the transmembrane and extracellular potentials, respectively; $\hat{\sigma}_i$ and $\hat{\sigma}_e$ are the intra- and extracellular conductivity tensors, respectively; I_m is the volume density of the transmembrane current, and I_0 is the volume density of the stimulation (shock) current.

The transmembrane current is the sum of the capacitive and ionic currents

$$I_m = \beta \left[C_m \frac{\partial V_m}{\partial t} + I_{\text{ion}}(V_m, t) \right]$$

where β is the surface-to-volume ratio, C_m is the specific membrane capacitance, and $I_{\text{ion}}(V_m, t)$ is the ionic current described by the Beeler-Reuter model of ion channel kinetics.

The passive parameters of the model, such as intra- and extracellular conductivities along and across the direction of the fibers and transmembrane capacitance, were the same as those used by Latimer and Roth (45). Their values are listed in Table 1. The boundary conditions for the extracellular potential for the upper facet of the slab were no flow (sealed)

$$\frac{\partial \phi_e}{\partial \bar{n}} = 0$$

where $\bar{n}$ is the unitary vector normal to the boundary. For the side facets, they were grounded (zero potential)

$$\phi_e = 0$$

and the boundary conditions for the lower facet were either zero potential (grounded) or no flow (35). A grounded boundary in simu-

lation corresponds to the tissue-bath interface with superconductive bath in experiment. A sealed (no flow) boundary corresponds to the heart surface exposed to air or any other nonconductive material (e.g., glass, fat) (23). We believe that such choice of the boundary conditions fostered the scroll wave to remain inside the slab, because the filaments were less likely to attach to grounded boundaries compared with sealed ones (see RESULTS). The intracellular space was considered sealed except for one simulation when it was grounded at the bottom. The boundary conditions are specified for every particular figure.

For the description of the ion channel kinetics, we employed the modified Beeler-Reuter model of ventricular action potential (71), which proved to be robust and stable for very high transmembrane voltages. Action potential during ventricular tachycardia is substantially shorter than a normal action potential (34). To account for that, we accelerated the slow inward calcium current by a factor of eight (71). This resulted in a reduction of the action potential duration to ~ 100 ms. In addition, we suppressed the conductance of the sodium channels by 1.5 times. Transmembrane voltages higher than 200 mV and lower than -200 mV were clipped to these extreme values as would occur in normal myocytes where at such voltages electroporation begins to play a role (18, 40). All of these modifications were obviously proarrhythmic and fostered induction of reentry that could be fit in a slab of a comparatively small size and hence, required less computational time.

The S1-S2 protocol of stimulation was used to induce reentry. An extracellular unipolar stimulating electrode ($0.7 \times 0.35 \times 0.35$ mm³) was positioned at the center of the upper layer of the slab. An initial cathodal S1 pulse (strength = -0.5 mA, duration = 2 ms) was delivered to resting tissue at $t = 0.2$ ms to generate an outwardly propagating wave. At $t = 88$ ms, we applied an S2 stimulus (strength = -3 mA, duration = 20 ms) through the same electrode, which initiated reentry via the process of break excitation. The range of the S1-S2 coupling intervals for which at least one rotation of reentry could be achieved was estimated to be from 79 to 95 ms for our S1 and S2 stimuli strengths. A similar protocol of VE-induced reentry has been used previously by Hildebrandt and Roth (37).

Because the slab-electrode configuration is symmetrical with respect to the midlines of the x - and y -axes and because fiber rotation with depth is not considered, one can make calculations only for a quarter of the slab. We used an explicit forward Euler algorithm to solve for V_m and biconjugate gradient method (61) with relative tolerance <0.001 to iteratively find the extracellular potential. To reduce the expense of the simulations, the equation for the extracellular potential was solved once every five time steps during the propagation of the wave and at every time step during application of the stimulus (1). For all simulations, we had the space step of 0.1 mm in the direction parallel to the fibers and 0.05 mm in the perpendicular direction (45); the time step was 10 μ s. By repeating some of the simulations for half the space and time steps, we confirmed the stability of our calculations and found $<0.1\%$ error for V_m . The calculations were performed on a Dell computer with a Pentium IV 2-GHz processor. The time required for 250 ms of simulation varied from 5 to 56 h depending on the size of the slab.

The basic electrophysiological parameters determined from the simulations were close to those reported previously by Roth and colleague (46, 65). For the tissue in the resting state, the length constant parallel to the fibers was 0.433 mm, the length constant perpendicular to the fibers was 0.173 mm, and the membrane time constant was 6.1 ms. Roth (66) emphasized large discrepancies in available experimental data on conductivity and length constant values for mammalian ventricular muscle, but found that most of the results agreed on the ratio of longitudinal to transverse constants, which was ~ 2.5 . In our simulations, the conduction velocity parallel to the fibers was 0.31 mm/ms, and the conduction velocity transverse to the fibers was 0.12 mm/ms. These numbers are lower than those usually reported for mammalian ventricular tissue: ~ 0.6 and ~ 0.2 mm/ms, respectively (62), even taking into account the interspecies

differences (21). The discrepancy is mainly due to reduced excitability of our tissue achieved through the suppression of the sodium channel conductance by 1.5 times. The same cause accounts for decreased maximal rate of the action potential upstroke that was 62 mV/ms. Slowing of conduction during arrhythmia or rapid pacing has been observed experimentally before (5, 14). We took advantage of this phenomenon to be able to implement the calculations in a slab of relatively small sizes.

3D volume data were analyzed in MATLAB (Mathworks) and visualized by using the isosurface approach. Depolarized regions with $V_m > 5$ mV are enclosed by a red isosurface, and blue isosurfaces contain hyperpolarized regions with V_m less than -90 mV. The wave front is shown as a green area in which the following two conditions hold: the value of the sodium m -gate exceeds 0.95 and the value of the h -gate is >0.05 . Therefore, the wavefront position somewhat corresponds to the region of high magnitudes of the depolarizing inward sodium current. The phase singularity filament (magenta) is found as the line of intersection of the two surfaces determining the wave front: $m = 0.95$ and $h = 0.05$. In this case, by definition, the filament is always at the edge of the wave front. A similar approach of phase singularity detection has been used previously in 2D models (22).

RESULTS

VE-Induced Reentry in a 3D Bidomain Slab of Cardiac Tissue

Figure 2 demonstrates the time course of the transmembrane potential distribution for the intramural reentry initiated by the S2 stimulus applied to the cardiac tissue in a relative refractory state after S1. The image for every frame shows only a quarter of the slab (corresponding to the distal quarter of the slab shown in Fig. 1). The complete picture of the transmembrane potential distribution can be imagined by reflecting the image symmetrically with respect to the x - and y -axes and the origin. This applies to all of the figures below.

At 13 ms after the application of S2, a pronounced characteristic pattern of stimulus-induced VE polarization is developed. Additional slight hyperpolarization (blue) at the back and bottom sides of the slab is associated with zero extracellular potential boundary conditions as indicated by the diagram on the right. The hyperpolarization at VAs (blue) deexcites the cells. The wave front is initiating from the depolarized VC area (red) at some distance from the electrode as shown by the arrows. It does not initially originate from the region in the immediate vicinity of the electrode, which has the highest value of the transmembrane potential gradient. This region does induce excitation but at a later stage when the strong hyperpolarization is decreased with time by intensive depolarization diffusion and the sodium m -gates are able to open.

At 29 ms, there are two main directions of the propagation: one corresponds to the part of the wave front created by the break excitation that rapidly spreads along the fibers and the other corresponds to the part of the wave front crawling over the bottom of the slab across the fibers. At 53 ms, the two parts of the propagating wave have met with the boundaries and are turning back into the area initially occupied by the VC. This area still has not reached the resting state; therefore, the propagation is substantially slowed down and the amplitude of the action potential is also decreased. At 77 ms, the small reentrant part of the survived wave front is appearing as a breakthrough and is moving toward the origin. The double rotation of the wave in the two perpendicular planes (x - y and

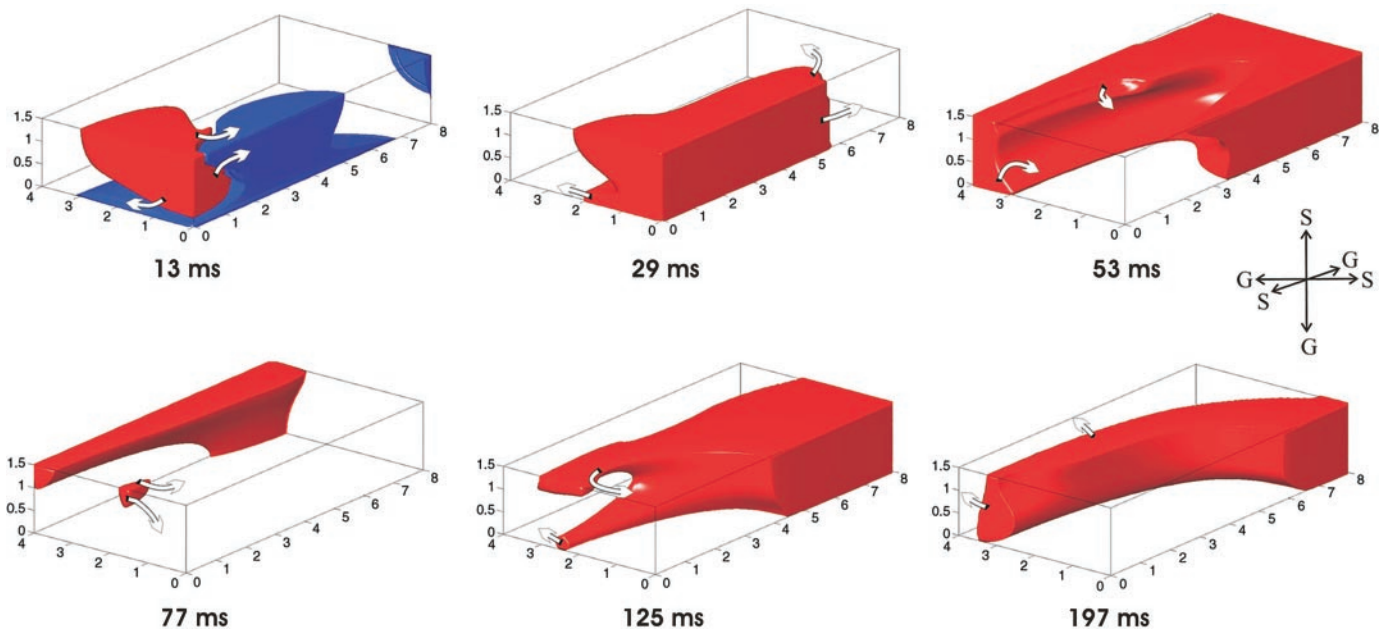


Fig. 2. Transmembrane potential (V_m) for intramural reentry at different times after S2 stimulus application. Only 1/4 of the slab is shown for every frame. The red depolarized region corresponds to $V_m < 5$ mV, and the blue hyperpolarized regions corresponds to V_m less than -90 mV. The arrows illustrate the direction of the propagation of the depolarization. The scheme on the right describes the boundary conditions for the 1/4 of the slab presented. S, boundary with sealed extracellular potential; G, boundary with grounded extracellular potential.

y - z) continues as shown at 125 ms. The reentry makes three cycles and is self-terminated by 197 ms.

Scroll Waves and Filaments for Intramural and Transmural Reentry

Figure 3 illustrates the dynamics of the intramural scroll wave front (green), the filament (magenta) for the same sim-

ulation, and the same frames as in Fig. 2. At 13 ms, one U-shaped wave front is generating at some distance from the electrode along the boundary between the VC and VA. Another weak O-shaped excitation wave appears above the lower hyperpolarized facet of the slab. The edges of the two wave fronts located in the x - z plane merge together resulting in one long U-shaped filament at 29 ms. Note that if the initial

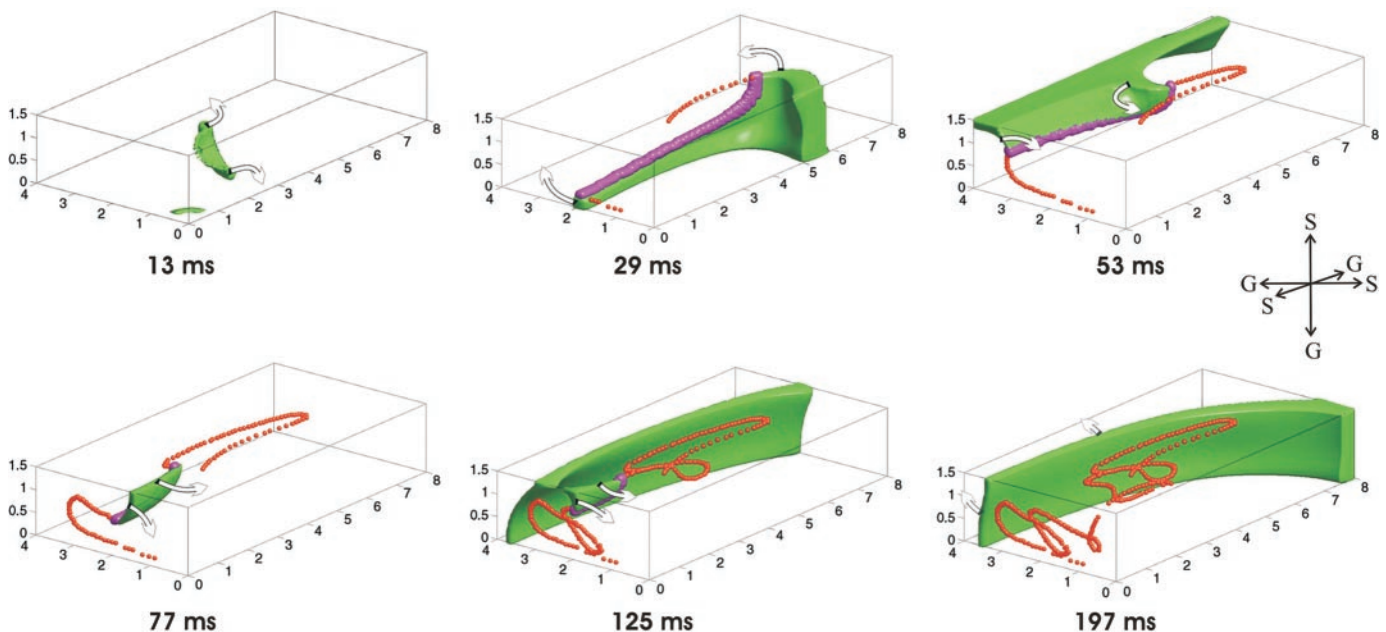


Fig. 3. Scroll wave (green) and the filament (magenta) for intramural reentry at different times. The scroll wave rotates around an L-shaped filament. The orange dots track the positions of the filament on the x - y and y - z planes. The arrows show the direction of the wave front propagation. The scheme on the right describes the boundary conditions for the 1/4 of the slab presented.

U-shaped filament was aligned in the plane transverse to the fibers direction, the one created after the transformation is aligned parallel to the direction of the fibers. The orange dots track the positions of the filament's ends (after the merging) on the upper x - y surface of the slab (epicardium) and on the cross section with the y - z plane (intramural plane) delineating the core of the reentry. One can see that the ends of the filament first move along nearly straight lines in each plane (29 ms) and then make sharp turns when they approach the boundaries (53 ms). During this process the U-shaped filament stretches and shrinks, with detachments of closed-phase singularity-free wave fronts from the newly reentrant scroll wave (125 ms). After one complete loop of reentry rotation, the filament goes inside the core of that loop for the second rotation, because the region inside the loop was not activated and therefore is more excitable than the rest of the tissue that did not have enough time to recover. While the filament is moving inside the loop, the tissue recovers and is ready for the next cycle of the reentry. Such behavior, however, leads to reduction of the size of the reentry core: every next loop is smaller than the previous one. Finally, when the ends of the filament come too close to each other, they annihilate (197 ms) and the reentry terminates. The last residual wave of excitation detached from the scroll wave is propagating outward (197 ms). The described dynamics of shrinking core seems to be true only for a slab of quite a small size, when the tissue does not have enough time to recover.

Figure 4 demonstrates the movement of the transmural I-shaped filament. The only difference from the previous simulation with intramural reentry is that the lower facet of the extracellular domain of the slab is sealed from the very beginning. In this case, there is no hyperpolarization at the bottom during the application of the S2 stimulus, and therefore, the second O-shaped filament is not created as in the previous

situation. The only U-shaped filament is induced by the VE polarization mechanism (16 ms), and its end on the x - z plane is moving downward, finally passing onto the lower surface of the slab. At this moment, one U-shaped filament is transformed into two I-shaped filaments (keep in mind that only one-fourth of the slab is shown) making three complete rotations until they collide and annihilate (232 ms). The same mechanism of shrinking of the core (described above) is responsible for the self-termination of the reentry.

Effect of the Slab Thickness on the Scroll-Wave Reentry

To find out how the thickness of the slab might affect the reentry type and its dynamics, we conducted simulations similar to those described in Figs. 3 and 4, except for the 0.5- and 3-mm-thick slabs. For one pair of the simulations, the extracellular potential at the lower facet was set to zero (Fig. 5) and for the other pair, the extracellular space was considered sealed at the bottom (Fig. 6). The horizontal dimensions of the slab, as well as the stimulus strength and the S1-S2 coupling interval were left unchanged.

In the 0.5-mm-thick slab with grounded extracellular potential at the bottom (Fig. 5A) the scroll-wave was generated in the same manner as for the 1.5-mm slab (Fig. 3). A weak O-shaped filament, created over the hyperpolarized bottom facet of the slab merged with a VE generated U-shaped filament, resulting in a U-shaped filament aligned in the plane parallel to the fibers (28 ms). The last moves transverse to the direction of the fibers (as illustrated by the orange dotted traces) until its lower part touches the bottom facet (34 ms), which transforms the U-shaped filament into two I-shaped ones (keep in mind that only one-fourth of the slab is shown in the figures). These two do not survive for long, but annihilate with each other by 43 ms and therefore terminate the scroll wave. Thus as opposed to the

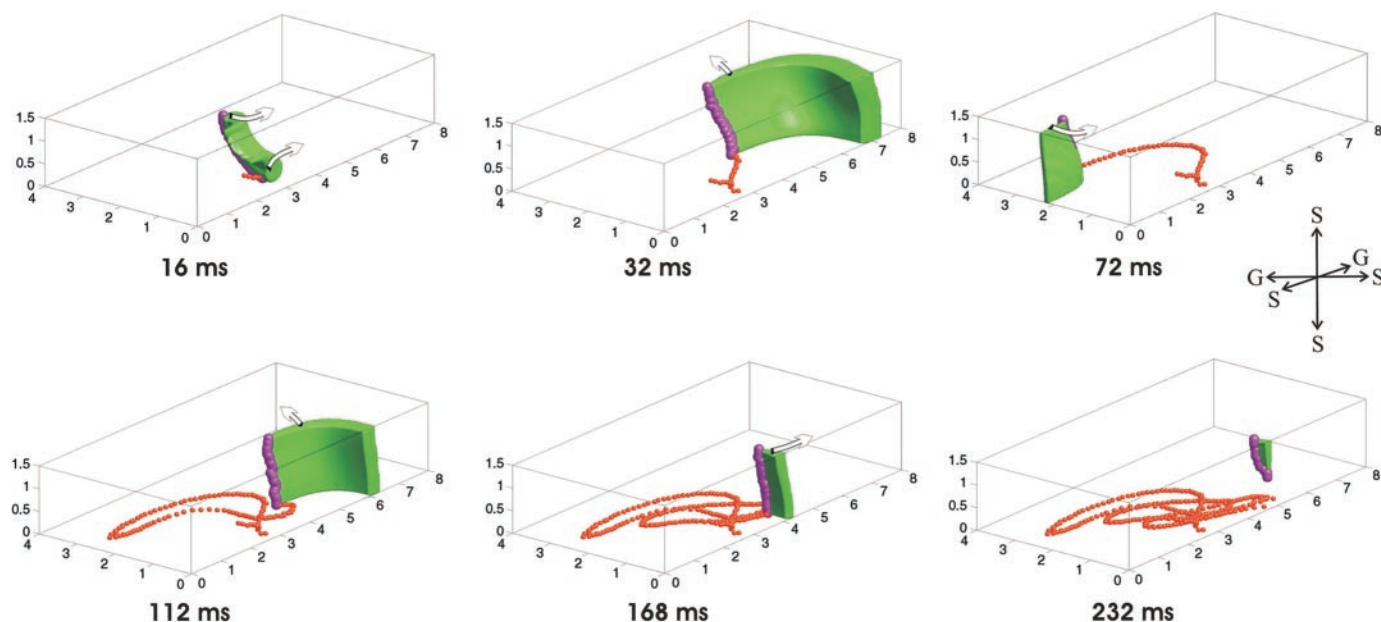


Fig. 4. The scroll wave (green) and the filament (magenta) for transmural reentry at different times. Reentry is obtained under the same conditions as in Fig. 3, except the lower facet of the slab is made grounded. In this case, after 16 ms the lower end of the filament moves to x - z plane instead of y - z as in Fig. 3. The orange dots track the positions of the filament on the x - y and x - z planes. The arrows show the direction of the wave front propagation. The scheme on the right describes the boundary conditions for the 1/4 of the slab presented.

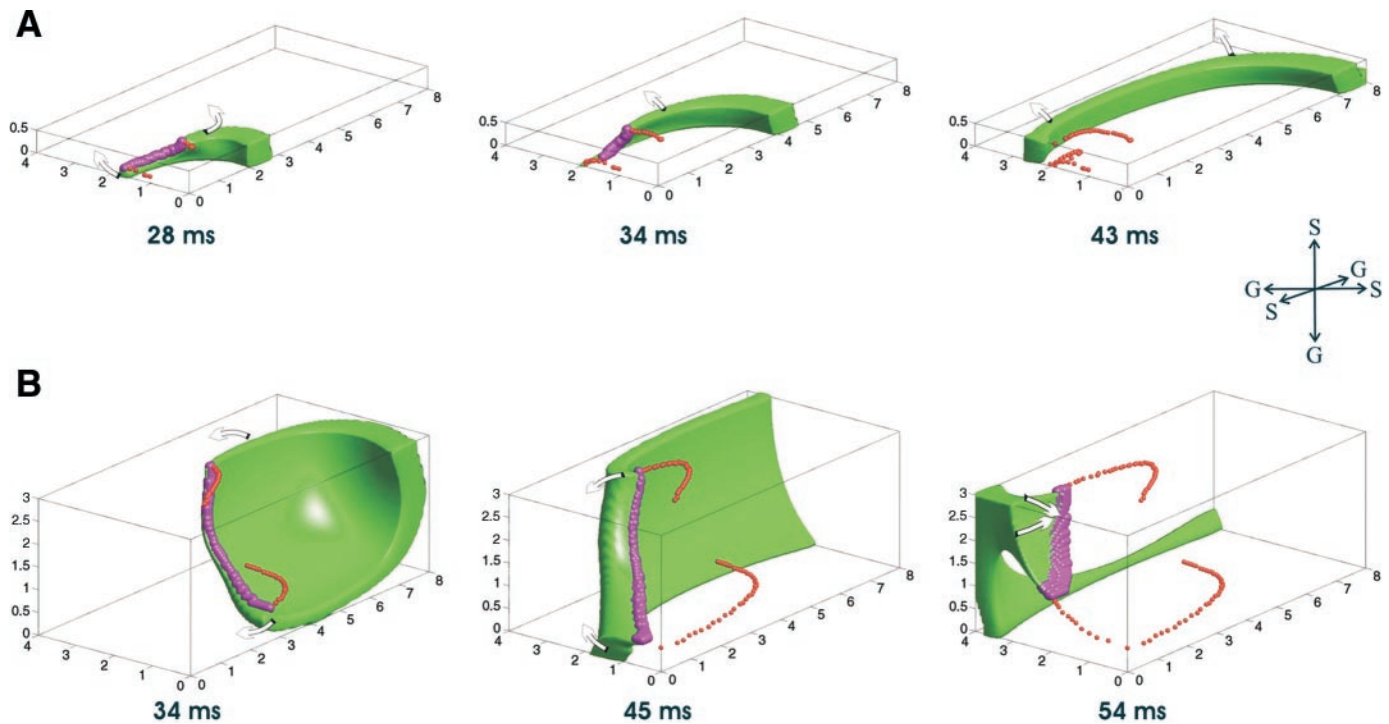


Fig. 5. The scroll wave (green) and the filament (magenta) for the slabs of 0.5- and 3.0-mm thickness with grounded extracellular potential at the lower facet. *A*: for the 0.5-mm slab, the S2 pulse resulted in a U-shaped filament moving transverse to the fibers. The orange dots track the positions of the filament on the x-y and y-z planes. At 34 ms, the U-shaped filament transforms into two I-shaped filaments, which annihilate with each other by 43 ms. *B*: for the 3.0-mm slab, the stimulus-induced U-shaped filament collapses into a spot at the upper surface after 54 ms. Arrows show direction of wave front propagation. The scheme on the right describes the boundary conditions for the 1/4 of the slab presented.

1.5-mm-thick slab, in the thinner slab, the scroll wave could not be sustained even for one full rotation.

Interestingly, in a 3.0-mm-thick slab with grounded extracellular potential at the lower boundary (Fig. 5*B*) the U-shaped scroll wave also did not survive. In this case, due to the larger distance between the electrode and the lower facet, initially no O-shaped filament was created over the bottom; the only U-shaped filament generated between the VC and VA first traveled away from the electrode along the direction of fibers and then at nearly 34 ms it made a sharp turn and retreated back until at 44 ms it transformed into another U-shaped filament moving already transverse to the fibers. The last gradually shrank and collapsed into a spot after 54 ms. The scroll wave self-terminated because the central area near the electrode did not have enough time to recover from the shock-induced depolarization.

Figure 6 illustrates scroll-wave dynamics for 0.5- and 3.0-mm-thick slabs when no current flow condition at the lower facet is imposed. In the first case of 0.5 mm, the filament is from the very beginning intramural and its dynamics are similar to that observed for the 1.5-mm slab: three consecutive loops with diminishing amplitude and annihilation. For the thick slab of 3 mm, a U-shaped filament is created at the boundary of VA and VC, but as it retreats back toward the electrode, its lower part touches the bottom of the slab, giving birth to a couple of I-shaped filaments, which in a while merge with their lower ends resulting in a U-shaped filament again. This behavior continues for two complete rotations until the filament shrinks to a spot at 160 ms. The series of I-to-U-to-I

transformations for this case can be better understood if one imagines a semi-infinite space with VE-induced U-shaped scroll wave cut by a plane at 3-mm depth beneath the surface. The smaller number of rotations compared with the 0.5-mm case is due to a stronger localization of the VE near the pacing electrode tip, and as a result, smaller amplitude of the first loop of the filament trajectory.

Effect of Boundary Conditions on the Filament Dynamics

From the results presented above, the following observation can be made. For the cases when the extracellular potential at the bottom of the slab was grounded, VEs induced only intramural filaments (Figs. 3 and 5), whereas the sealed extracellular space at the bottom lead to transmural reentry as well as intramural (Figs. 4 and 6). What was the role of the boundary conditions in determination of the type of reentry? One possible argument is that the region near the boundary for the cases with grounded extracellular potential experienced hyperpolarization during the application of the S2 stimulus (see Fig. 2) and therefore could not initially anchor a filament, which, as known, must be generated at the borderline between the VC and VA. However, even later during the whole time course of the intramural reentry (Fig. 3), the lower end of the filament never passes on to the bottom facet of the slab (with the exception of the preannihilation stage for the 0.5-mm case in Fig. 5) as opposed to the situation with the sealed extracellular space in Figs. 4 and 6.

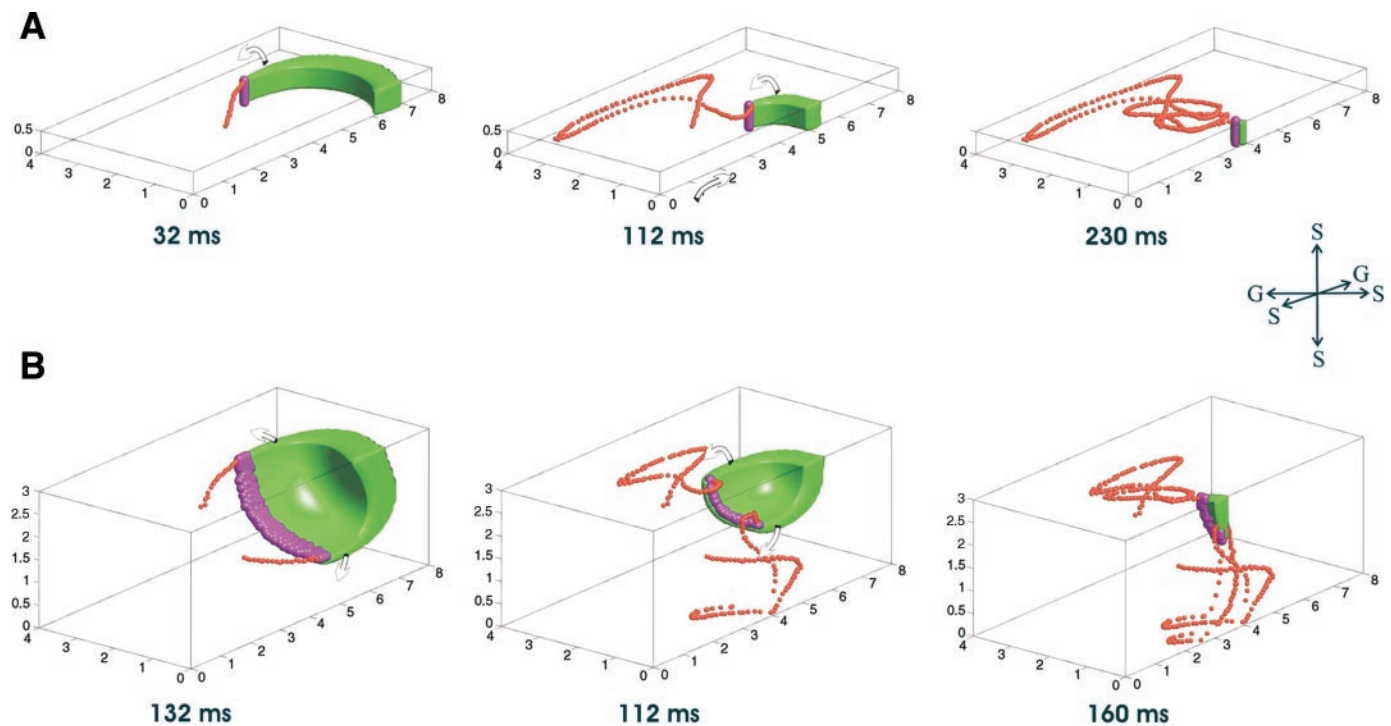


Fig. 6. Scroll wave (green) and the filament (magenta) for the slabs of 0.5- and 3.0-mm thickness with sealed extracellular space at the lower facet. *A*: for the 0.5-mm slab, the I-shaped filament meanders along loop-like trajectories (as shown by the orange dots on the *x-y* plane) making 3 rotations before annihilation at 230 ms. *B*: for the 3.0-mm slab, the filament topology alternates between I- and U-shaped. Arrows show the direction of the wave front propagation. The scheme on the *right* describes the boundary conditions for the 1/4 of the slab presented.

Effect of tissue-bath interface on scroll dynamics. To simplify the explanation below by grounded or sealed boundary, we shall imply a boundary with grounded (zero) or sealed (no flow) boundary conditions for the extracellular potential, respectively. For the real heart, a grounded boundary can be considered to be a certain degree equivalent to the tissue-bath interface, and a sealed boundary can represent the tissue-air interface.

In an attempt to elucidate the effect of the boundary conditions on the scroll-wave dynamics, we started the simulation with the sealed bottom facet of the slab, the same as shown on Fig. 4. At 82 ms after the application of the S2 pulse, we then rapidly grounded the bottom. Figure 7 demonstrates the results with the blue trace representing the movement of the filament's lower end after the change of the boundary conditions and the orange trace, the dynamics if the change did not occur. It can be seen that the blue trajectory of the filament with the grounded bottom of the slab somewhat deviates from the initial "sealed" orange trajectory, especially when the filament approaches the right (sealed because of the symmetry) *x-z* plane. At 102 ms the blue trace detaches from the bottom and climbs up on the *x-z* plane. In the complete picture, this means that the two I-shaped filaments merged with their lower edges and created one U-shaped filament. However, this does not last long; the filament returns back to the bottom plane and makes a loop approximately after the orange trace of the original filament (130 ms). Nevertheless, at 178 ms, the filament detaches from the bottom for the second time. This time, the bottom end of it climbs up the *x-z* plane higher and the filament dies out. The frames for 194 and 231 ms show that had the

bottom of the slab remained sealed, the original transmural filament would make one more (the third) rotation before the reentry termination.

The described dynamics leads us to the conclusion that the scroll-wave filaments tend to be repelled from boundaries with grounded extracellular potential compared with the sealed boundaries. In other words, our statement implies that a scroll-wave filament would attach to the surface of the heart that is exposed to air rather than to bath.

Effect of surface electroporation on scroll dynamics. One can go further and change the boundary conditions from sealed to grounded for both extracellular and intracellular potentials. Such modification would essentially mean that the transmembrane potential at the bottom of the slab is made equal to zero and would correspond to the situation of tissue-bath interface with electroporated cells. It is natural to expect that a scroll-wave filament would detach from the electroporated and therefore depolarized boundary. Figure 8 illustrates the results of the simulation started with sealed extracellular and intracellular potentials at the bottom facet of the slab (as in Fig. 4), both of which are then rapidly grounded at 82 ms after the application of the S2. The scroll wave and the filament immediately detach from the bottom of the slab and jump onto the *x-z* plane (83 ms). The filament thus undergoes the I to U transformation. The lower end of the filament travels along the whole perimeter of the slab over the depolarized part of the slab, whereas the upper end makes one loop on the upper surface. The speed of the rotation of the reentry is quite quick, and the effective size of the excitable tissue is reduced by the electroporation-caused depolarization, so the reentry dies after one rotation (170 ms).

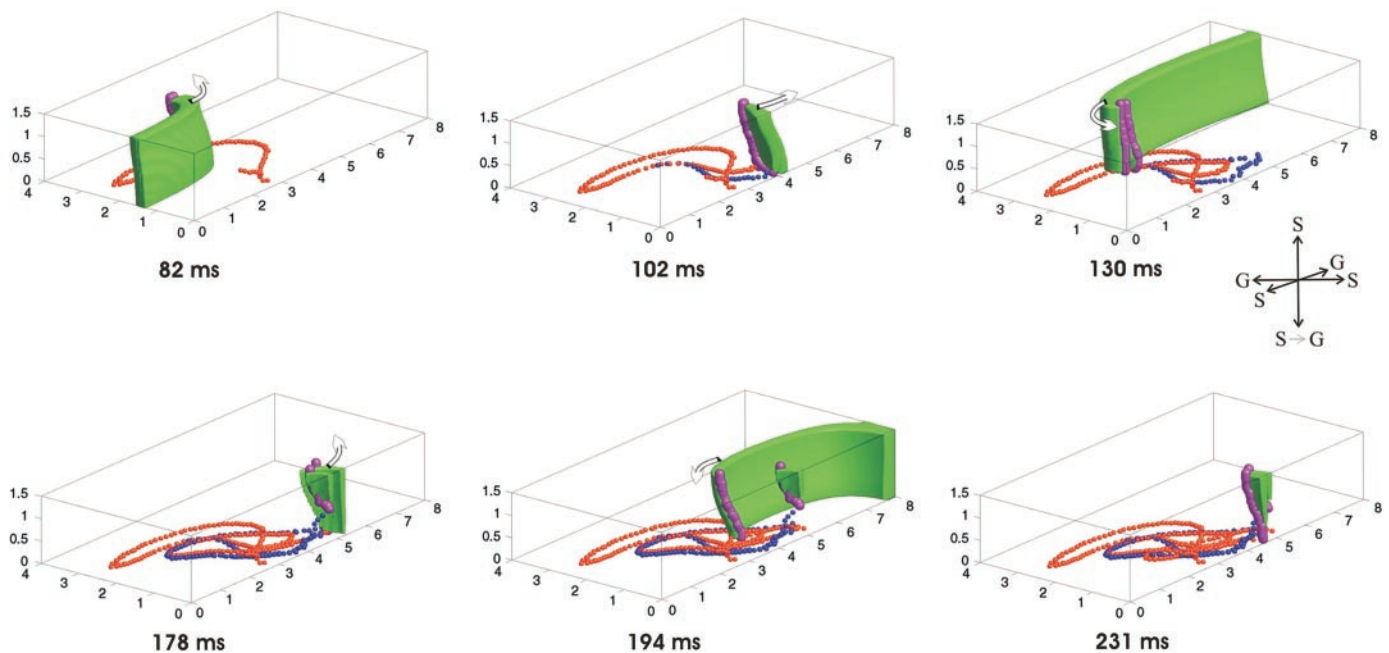


Fig. 7. The scroll wave (green) and the filament (magenta) at different times, illustrating the repulsion of the filament from the grounded toward the sealed boundary. Results of the 2 simulations are superimposed. One simulation is performed with the sealed extracellular potential for the bottom facet for the slab with the orange dots demonstrating the trajectory of the lower end of the filament. The other simulation is done with the potential suddenly changed from sealed to grounded at 83 ms after the S2 application. The blue dots track the positions of the filament for this case. In the second case, the filament end moves closer to the x - z plane and dies out after 2 rotations. The arrows show the direction of the wave front propagation. The scheme on the right describes the boundary conditions for the 1/4 of the slab presented.

The simulation confirms the fact that a scroll-wave filament cannot attach to an electroporated boundary.

DISCUSSION

3D active bidomain models of cardiac muscle have not been used to study functional reentry until very recently due to their significant computational expense (74). Roth and colleague (37, 67) investigated the protective effect of stimulation in a myocardial tissue with cylindrical symmetry which calculation-wise reduced the problem to 2D. Preliminary data showing functional reentry in a model of a rabbit ventricle are presented by Aguel et al. (1).

In our work, we employed the active 3D bidomain model to demonstrate for the first time the possibility of inducing transmural and intramural scroll-wave reentry by the mechanism of VE-induced-phase singularity. This mechanism, in particular S1-S2 pacing protocol, has been used for the quatrefoil reentry (47) induction before and has been extensively investigated for 2D bidomain (37, 67).

For the purpose of determination of the position of the scroll-wave filament, we employed the method of intersection of two independent variables of the model, namely we chose m - and h -gates of the sodium channels as the most natural indicators of activation. In the BRDR model of ventricular action potential, excitation is triggered by the inflow of sodium ions, which starts on activation of the m -gate and ends after inactivation of the h -gate. Therefore, in our simulations, the wavefront was viewed as a region with inward depolarizing current and the filament was found as the edge of this wavefront. This approach does not rely on the choice of the phase plane origin or time delay as in the phase angle method (13, 39)

and, in our opinion, is more suitable in application to numerical simulations when all the variables are available. However, other methods of phase singularity lines detection on the basis of topological charge density calculation have been also developed recently (13, 39). Although there has been no work done comparing these methods, we believe that the possible minor discrepancy is not significant for the validity of our results.

Visualization of the scroll-wave filament reveals existence of the main topological types of the filaments predicted theoretically (49, 85) and obtained earlier in active monodomain simulations (57) as well as in passive bidomain simulations (69). Relative stability of the filaments, which is not always observed in other simulations (29, 36) may be attributed to the simple straight fiber geometry assumed in the model. Incorporation of the fiber rotation with depth into the model leads to earlier termination of the reentry for our settings and protocol of stimulation. This is illustrated by Fig. 9, which shows evolution of S1-S2 stimulus-induced scroll waves in $8 \times 16 \times 1.5$ -mm slab with a grounded lower boundary. All the conditions are the same as those for the Fig. 3 except that the fibers rotate equably with depth by a 90° angle from the top to the bottom of the slab. As a result, the wave front (green) and the filaments (magenta) are strongly twisted in the direction after that of the fiber rotation (28 ms). This distortion draws the ends of the filaments toward each other and terminates the quatrefoil scroll by 36 ms after the S2 application. Such an outcome is generally in agreement with the predictions of Berenfeld et al. (8, 9), who stated that filaments tend to align along the fibers, as well as with the results of Fenton and Karma (29), demonstrating a possibility of the filament destabilization by fiber

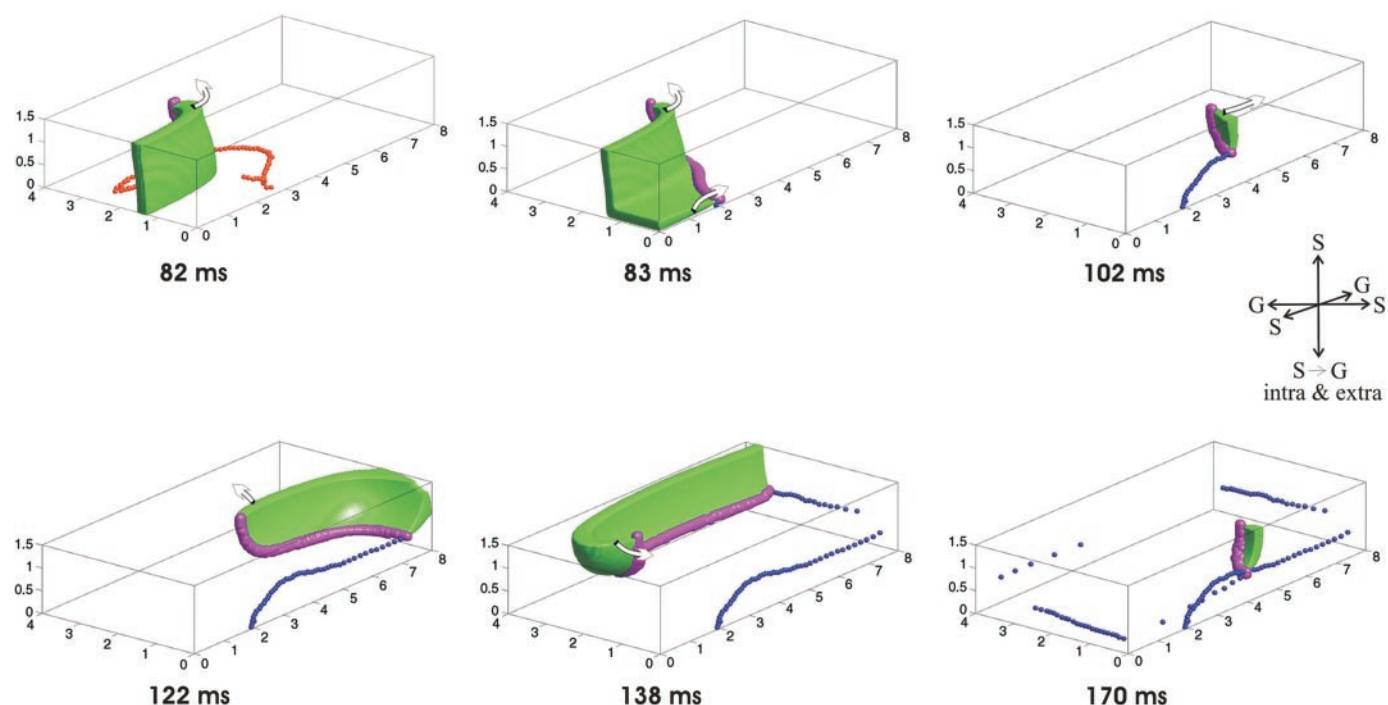


Fig. 8. The scroll wave (green) and the filament (magenta) at different times illustrating the detachment of the filament from the electroporated boundary. The simulation is started with the no-flow boundary conditions for both extracellular and intracellular potentials at the bottom facet of the slab. At 83 ms after the S2 application, the boundary conditions for the both potentials are altered to grounded, which causes immediate detachment of the lower end of the filament from the bottom. The subsequent trajectory of it is traced by the blue dots. The arrows show the direction of the wave front propagation. The scheme on the right describes the boundary conditions for the 1/4 of the slab presented.

rotation. Unfortunately, the model with fiber rotation does not possess the symmetry of the slab with respect to the stimulating electrode and therefore computationally is four times more expensive than that without the rotation. This fact on this stage prevented us from conducting a more detailed and comprehensive investigation of the fiber rotation effect using the bidomain model.

Dimensions of the slab along with spiral tip trajectory strongly determine the evolution of the scroll-wave reentry. In our simulations, the reentry was sustained at most for three rotations and then self terminated, which we attribute to the small size of the slab, although other factors such as the S2 stimulus strength or restitution properties of the tissue might

have played an important role. Figure 10 demonstrates phase singularity traces for the 2D bidomain simulations conducted with the same protocol and the same parameters as those used in our 3D bidomain. Only quarters of 16×8 and 24×12 mm rectangular domains are shown. In the first case of a 16×8 mm rectangle, the reentry terminates after the second rotation (Fig. 10, *bottom*), whereas for the bigger rectangle it completes more than three cycles (Fig. 10, *top*). Previously, for 2D models both nonsustained (67) and sustained (37) reentries have been obtained by using S1-S2 protocol of stimulation.

As can be seen from Figs. 3, 6, and 10, in the beginning of the reentry the surface ends of the scroll-wave filaments meander along elongated oval trajectories. Similar linear reentry

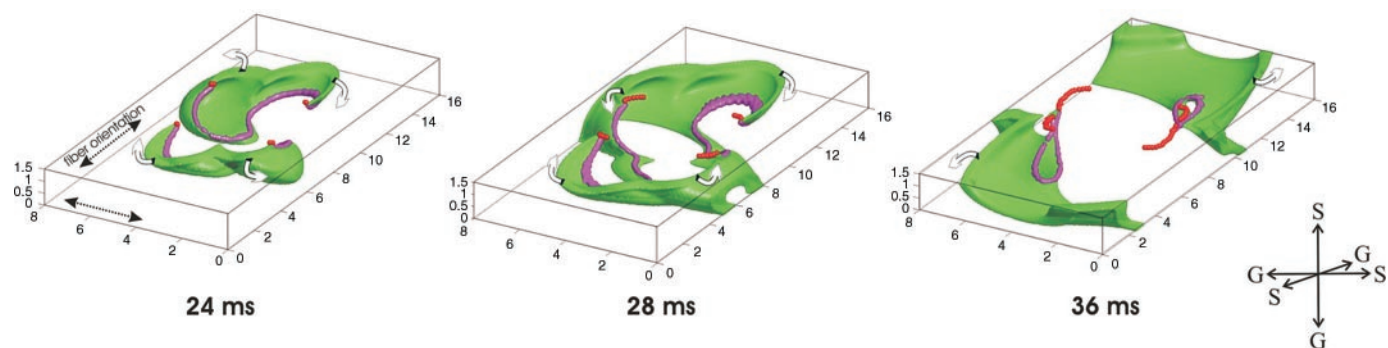


Fig. 9. The scroll wave (green) and the filament (magenta) at different times for the bidomain model with fiber rotation. Direction of the fibers rotates by 90° from the upper to lower facet of the slab. Reentry self terminates before completion of the first cycle. Arrows show the direction of the wave front propagation. The orange dots track the positions of the filament on the upper x-y surface of the slab. The scheme on the right describes the boundary conditions for the slab.

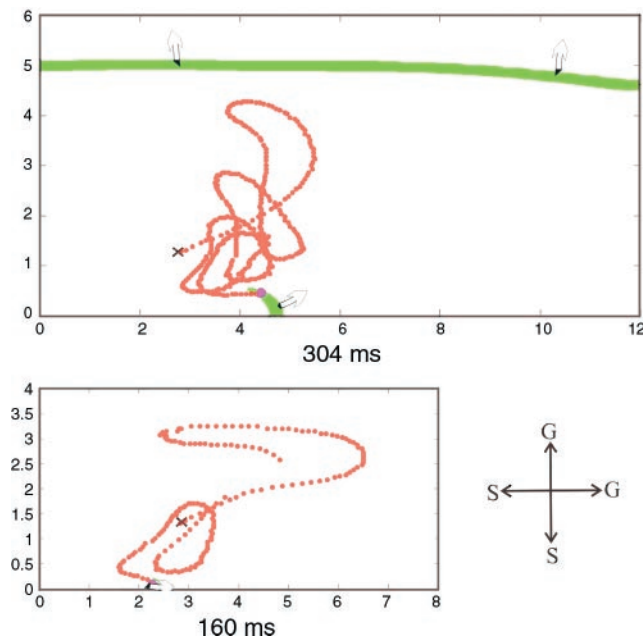


Fig. 10. Phase singularity trajectories (orange dots) obtained from 2D bidomain simulations for rectangular domains of 24×12 mm (top) and 16×8 mm (bottom) size with the same S1-S2 pacing protocol. Wave front of excitation is shown in green at 304 ms (top) and 160 ms (bottom) after the application of the S2 pulse. For the larger rectangle, the reentry is sustained for >3 rotations, whereas for the small rectangle, it terminated after the second rotation. Arrows show the direction of the wave front propagation. The scheme on the right describes the boundary conditions for the 1/4 of the domain presented.

cores have been observed in experiments at the surface of ventricular muscle (17, 22) as well as in monodomain 2D simulations with normal action potential (26, 29, 39). Because of the small size of the slab, with each rotation there is less and less excitable tissue left and every next loop of meandering becomes smaller and more circular compared with the previous one. This observation is consistent with the results of Efimov et al. (26), who achieved shrinking and circularization of the phase singularity trajectories by suppression of excitability through decrease in sodium channel conductance. The filaments disappear at the end of the reentry by the process of annihilation in a similar fashion as demonstrated previously (29, 53) for 3D monodomain.

One of the main advantages of the bidomain model over the cable equation-based models is that it gives an opportunity to correctly investigate the electrophysiological role of the boundary conditions for the intra- and extracellular potentials. Because most of the experimental studies of reentry using optical mapping are done on the hearts whose ventricular walls are either pressed to the glass of the chamber, hang in the air, or hang in the bath, we considered two types of boundary conditions for the extracellular potential, no-flow (sealed) and zero-potential (grounded). A grounded boundary in simulation corresponds to the tissue-bath interface with superconductive bath in experiment. A sealed (no flow) boundary corresponds to the heart surface exposed to air or any other nonconductive material (e.g., glass, fat) (23). In our work, we present computer simulations that provide evidence for repulsion of the scroll-wave filaments from the grounded boundaries (tissue-bath interface) toward the sealed ones (tissue-air interface) (Fig. 7). We believe that this effect is mainly due to the leakage of the

charge through the boundary, which makes it energetically more favorable for the filament to attach to a sealed boundary, in which there is no leakage at all. We are not aware of any experimental or theoretical work that would directly support or disprove this statement. Latimer and Roth (46) and Entcheva et al. (27) investigated the effect of bath on VEs, but did not address reentry induction.

Having the horizontal dimensions of the slab fixed, we varied the vertical dimension and found that for the case of sealed extracellular space at the lower boundary, the thin (0.5 mm) slab exhibited transmural I-shaped scroll wave, whereas in the slab of 3-mm thickness, the filament topology alternated between I- and U-shaped (Fig. 6). These observations suggest that a sealed boundary does not affect the filament dynamics strongly or, in other words, the reentry would not look very different if there was no such boundary at all and the slab was semi-infinite. Nevertheless, presence of the boundary plays a big role on the localization of the stimulus-induced VE polarization and therefore determines initial location of the VE-induced-phase singularities. This is why in the 0.5-mm-thick slab, the reentry was maintained for three rotations compared with two rotations in the thicker 3-mm slab (Fig. 6).

For the simulations with grounded bottom, boundary break excitation generated only a intramural scroll wave with a U-shaped filament that survived for the 1.5-mm-thick slab (Fig. 3), but terminated for the 0.5- and 3-mm-thick slabs before completing the first rotation (Fig. 5). For the 0.5-mm slab, its thickness was insufficient to sustain intramural scroll-wave rotations, whereas in the 3-mm slab, the bath was too far from the pacing site to allow for a quick drain of the stimulus-induced depolarization and timely recovery of the tissue in that central area. The hyperpolarizing effect of the tissue-bath interface due to the leakage of the extracellular current into the bath has been found before (27, 46). In our study, we demonstrate that 1) presence of the bath influences the topology of the VE generated scroll-wave filament. In particular, for our situation it led to creation of U-shaped filaments associated with intramural reentry, and 2) proximity of the tissue-bath interface helps to sustain the intramural reentry by fostering the recovery of the tissue near that interface.

Our results predict that the different boundary conditions may have a significant influence on the outcome of the defibrillation or fibrillation experiments. We admit that the superconductive bath and the grounded boundary conditions represent an idealized situation, the “zero-order” approximation. In reality, a bath has a finite conductivity, although quite large compared with that of the tissue. Nevertheless, from the considerations of continuity, we believe that our general conclusions about the prointra-mural reentry effect of the bath and the repulsion of the filament end from the tissue-bath interface in favor of the tissue-air interface will still be valid.

In our simulation, we also were able to confirm the fact that a scroll-wave filament cannot stick to an electroporated boundary. The antiarrhythmic effect of electroporation has been recognized before. Experiments of Al-Khadra et al. (2) demonstrated reduction of the heart vulnerability to external shocks. Simulations using 2D bidomain (4) suggested the explanation of the antiarrhythmic effect. Electroporation at hyperpolarized virtual anode areas creates break excitation, which reduces the size of the tissue susceptible to the reentry. Our simulations propose a different view at essentially the

same phenomenon but in 3D. Electroporation reduces the effective surface area to which the filaments can attach and therefore increases the probability of their annihilation in the new more congested space.

It must be mentioned that we used a very primitive model of electroporation by simply setting the transmembrane potential at the lower facet of the slab to zero. In reality, however, electroporation is a complex dynamic process consisting in the creation of sarcolemmal pores in response to transmembrane voltage higher than certain thresholds (200–600 mV) (19, 20). Electrophysiologically it results in reduced excitability, weak action potentials, and a highly elevated diastolic resting potential (2, 52). In the idealized extreme case of very strong electroporation, which we considered, there would be no action potentials at all, and the resting potential would be close to 0 mV. We suppose that incorporation of a more realistic model of electroporation may have resulted in a delayed detachment of the filament from the lower facet or no detachment at all.

An important limitation to our study is associated with the modification of the ion channel kinetics with the purpose to fit the reentry in a slab of comparatively small size and therefore reduce the computational expense of the simulations. Similar modifications have been extensively exploited in previous works (37, 48, 71). As a result, the core of the reentry becomes smaller and less elongated, and the reentry cycle length is also decreased (26). We suppose that having a more realistic action potential model would eliminate these effects and lead to a larger linear core with slower rotation of the spiral waves. The initial location, geometry, and topological transitions of the filaments would be affected too, depending on the temporal and spatial scales of the simulations. Nevertheless, we believe that our observations of different types of filaments for different boundary conditions would remain qualitatively unaltered, because the distribution of depolarized and hyperpolarized areas during the S2 stimulus, which determines the reentry type, does not depend on a particular action potential model.

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